

File permissions in Linux

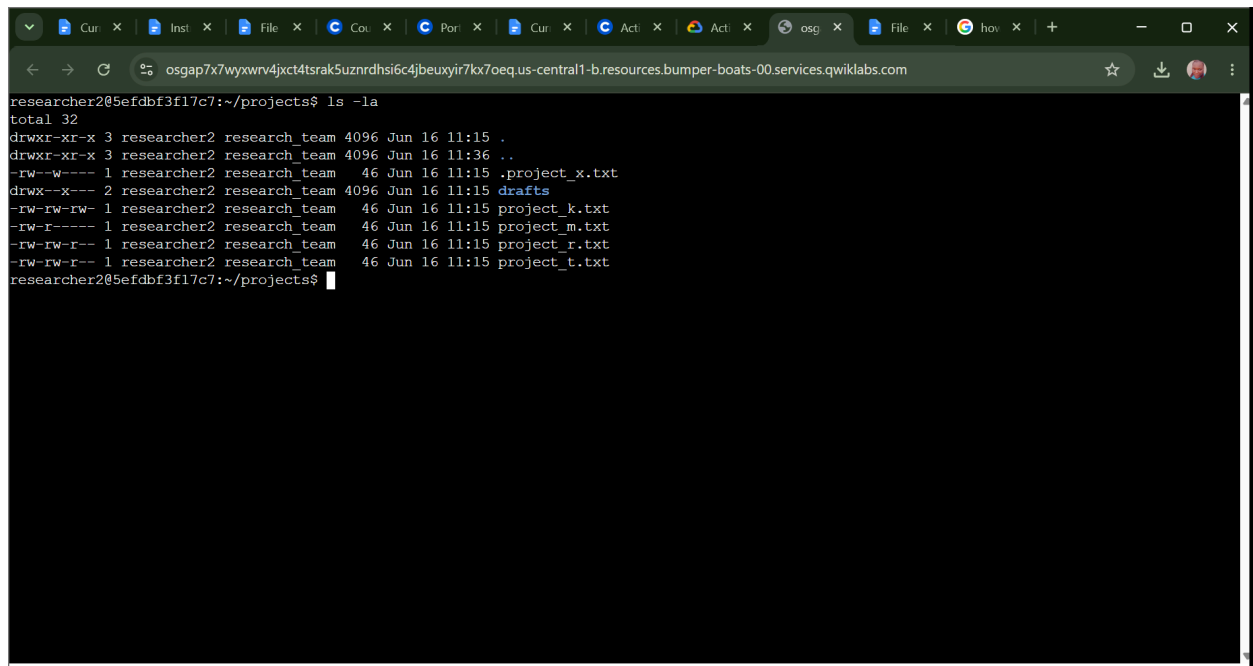
Project description

[As a cyber security professional working with the research team at a large organization, I audited files and directory permission in /home/reseachers2/projects. My goal was to ensure users have only the appropriate permission needed for their role .I examined existing permission and identified unauthorized access and used linux chmod command to modify permission to match the principle of least privilege.]

Check file and directory details

[To display all permissions, including hidden files, I used the command:

ls -la This command lists all files and subdirectories in the long format, showing the 10-character permission string for each item. The -a flag ensures hidden files like .project_x.txt are included.



```
researcher2@5efdbf3f17c7:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 16 11:15 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 16 11:36 ..
-rw--w---- 1 researcher2 research_team  46 Jun 16 11:15 .project_x.txt
drwx--X--- 2 researcher2 research_team 4096 Jun 16 11:15 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 16 11:15 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 16 11:15 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 16 11:15 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 16 11:15 project_t.txt
researcher2@5efdbf3f17c7:~/projects$
```

]

Describe the permissions string

[Choose one example: -rw-rw-r-- for project_k.txt Description for your template:

The 10-character string `-rw-rw-r--` represents the file type and permissions for `project_k.txt`.

Character 1 `-`: Indicates this is a regular file. `d` would mean directory.

Characters 2-4 `rw-`: The user, or file owner, has read and write permissions but no execute permission.

Characters 5-7 `rw-`: The group has read and write permissions but no execute permission. Characters

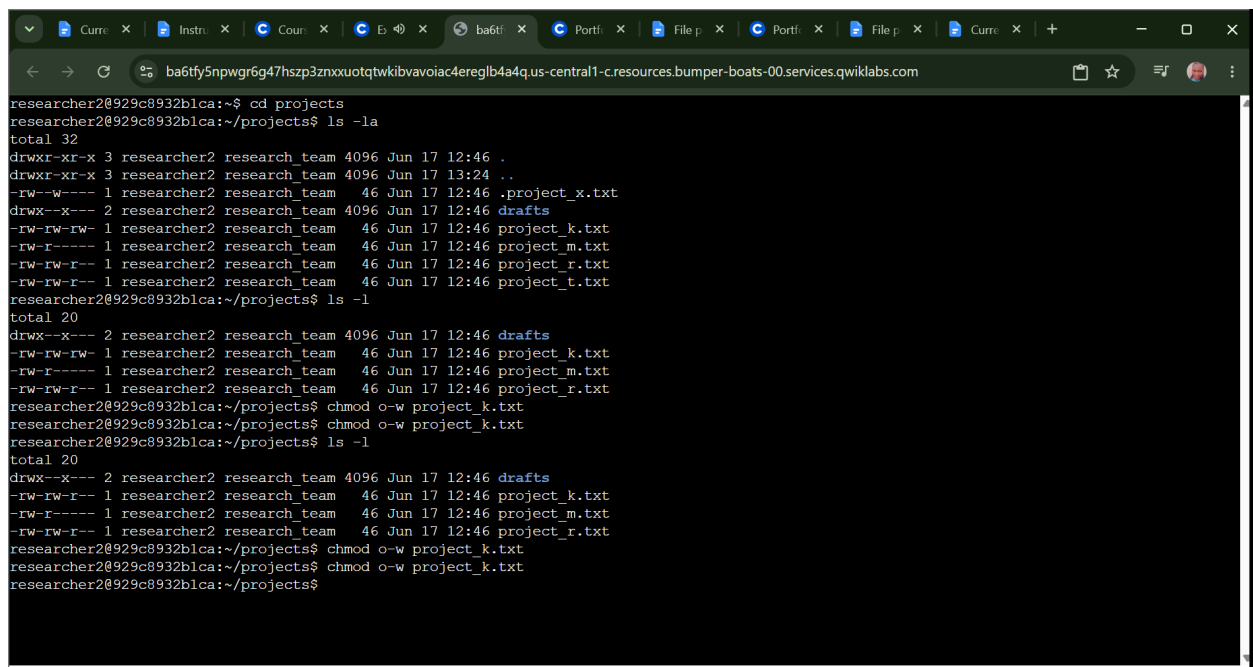
8-10 `r--`: Other, meaning all users not in the owner or group, has read permission only.

Security concern for this example: As the security professional for the research team, the `r--` for others violates the organization's policy. Only authorized research team members should access this file, so others should have no permissions.]

Change file permissions

[The organization does not allow others to have write access to any files. Analysis of current permissions:

From Step 3, none of the files currently give `w` to others. However, `project_k.txt`, `project_r.txt`, and `project_t.txt` give `r` to others, which also violates least-privilege for the research team. Linux commands to modify permissions: To remove other read access from the files that have it.

A screenshot of a terminal window with a dark background. The terminal shows a series of commands and their outputs. The user is in the `projects` directory. The first command is `ls -la`, which lists files with their permissions, owner, group, size, and date. The files listed are `.`, `..`, `project_x.txt`, `drafts`, `project_k.txt`, `project_m.txt`, `project_r.txt`, and `project_t.txt`. The permissions for `project_k.txt`, `project_r.txt`, and `project_t.txt` are `-rw-rw-r--`. The user then runs `chmod o-w project_k.txt` to remove write permission for others. This command is repeated for `project_r.txt` and `project_t.txt`. Finally, the user runs `ls -l` to show the updated permissions, which are now `-rw-rw----` for `project_k.txt`, `project_r.txt`, and `project_t.txt`.

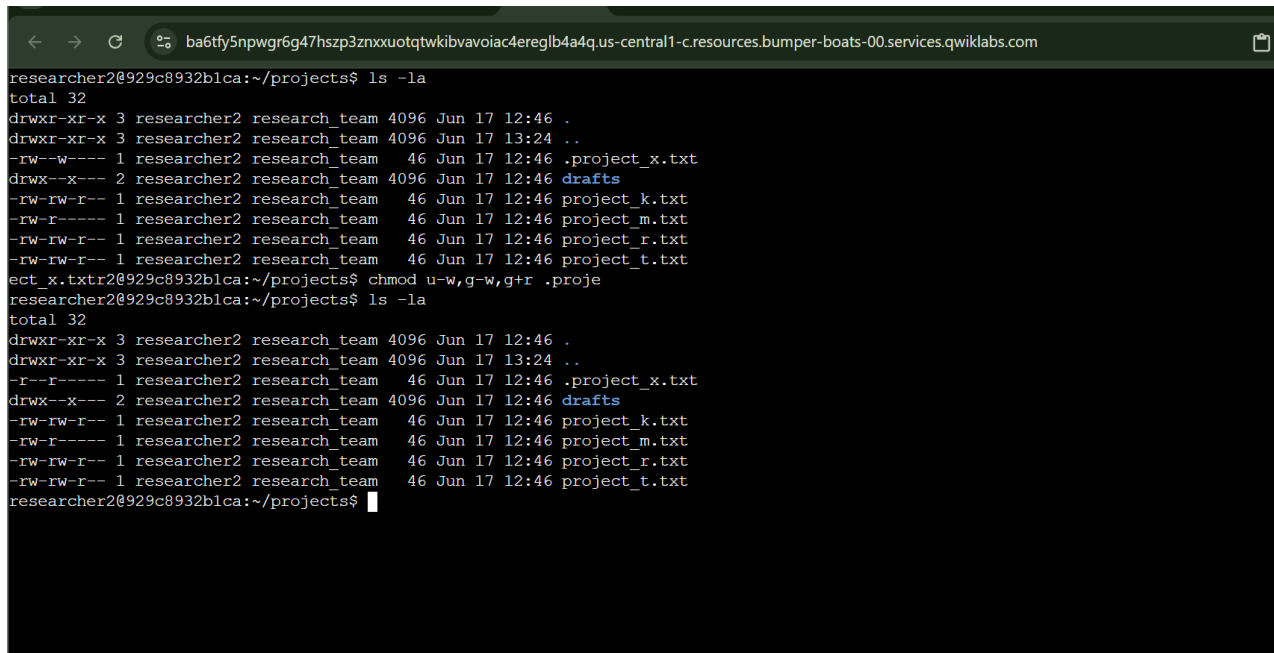
```
researcher2@929c8932b1ca:~$ cd projects
researcher2@929c8932b1ca:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 17 12:46 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 17 13:24 ..
-rw--w---- 1 researcher2 research_team 46 Jun 17 12:46 project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 17 12:46 drafts
-rw-rw-rw- 1 researcher2 research_team 46 Jun 17 12:46 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Jun 17 12:46 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jun 17 12:46 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jun 17 12:46 project_t.txt
researcher2@929c8932b1ca:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 17 12:46 drafts
-rw-rw-rw- 1 researcher2 research_team 46 Jun 17 12:46 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Jun 17 12:46 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jun 17 12:46 project_r.txt
researcher2@929c8932b1ca:~/projects$ chmod o-w project_k.txt
researcher2@929c8932b1ca:~/projects$ chmod o-w project_r.txt
researcher2@929c8932b1ca:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 17 12:46 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Jun 17 12:46 project_k.txt
-rw-r----- 1 researcher2 research_team 46 Jun 17 12:46 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jun 17 12:46 project_r.txt
researcher2@929c8932b1ca:~/projects$ chmod o-w project_k.txt
researcher2@929c8932b1ca:~/projects$ chmod o-w project_r.txt
researcher2@929c8932b1ca:~/projects$
```

This ensures only the owner and authorized group members can access the files.]

Change file permissions on a hidden file

[Hidden file from Step 3: `.project_x.txt` with permissions `-rw--w----` Breakdown: user=`rw-`, group=`-w-`, other=`----` Security issue: The group has write but no read permission, which is

unusual and violates least-privilege. The research team policy likely requires only the owner to access hidden files. Linux command to modify:

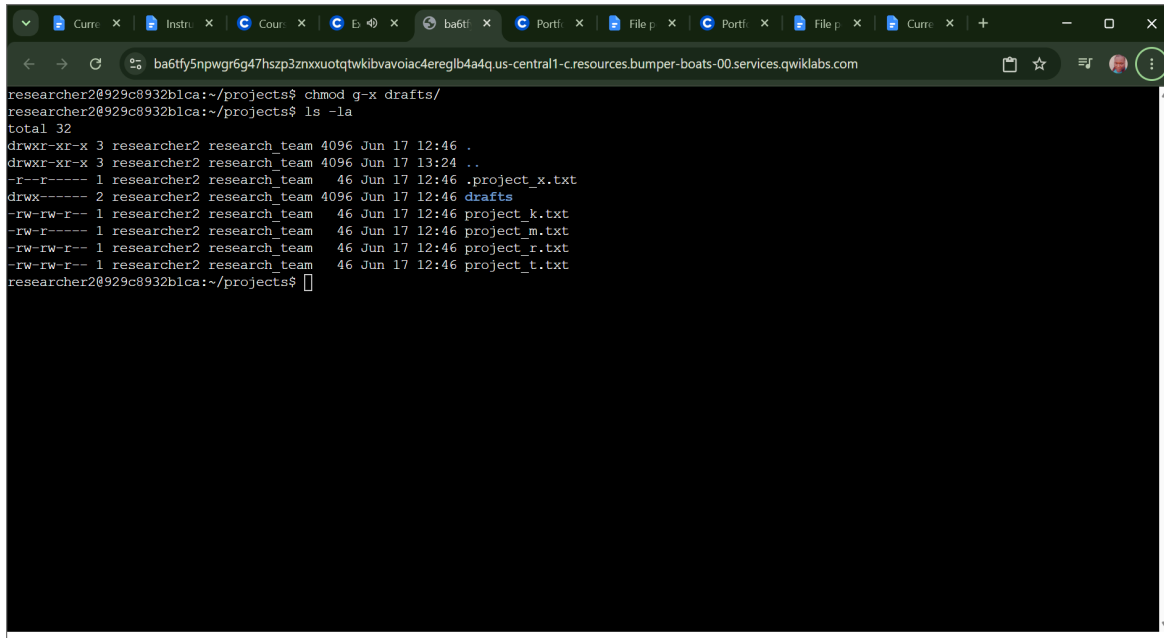
A terminal window with a dark background. The browser address bar at the top shows a URL starting with 'ba6tfy5npwgr6g47hszp3znnxuotqtwkibvavoia4ereglb4a4q.us-central1-c.resources.bumper-boats-00.services.qwiklabs.com'. The terminal shows a user 'researcher2' at a prompt 'researcher2@929c8932blca:~/projects\$'. They run 'ls -la' showing a directory listing with permissions, owner, group, size, date, and file names. The files include '.', '..', '.project_x.txt', 'drafts', 'project_k.txt', 'project_m.txt', 'project_r.txt', and 'project_t.txt'. Then they run 'chmod u-w,g-w,g+r .project_x.txt'. Finally, they run 'ls -la' again, showing the updated permissions for '.project_x.txt' as '-r--r-----'.

```
researcher2@929c8932blca:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 17 12:46 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 17 13:24 ..
-rw--w---- 1 researcher2 research_team  46 Jun 17 12:46 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 17 12:46 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 17 12:46 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 17 12:46 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 17 12:46 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 17 12:46 project_t.txt
researcher2@929c8932blca:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@929c8932blca:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 17 12:46 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 17 13:24 ..
-r--r----- 1 researcher2 research_team  46 Jun 17 12:46 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 17 12:46 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 17 12:46 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 17 12:46 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 17 12:46 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 17 12:46 project_t.txt
researcher2@929c8932blca:~/projects$
```

I removed write permissions from both user and group since the file is archived. I then added read permission for the group so both the owner and research team group can read the archived file, but no one can modify it. This matches the requirement that archived files are read-only for authorized users..]

Change directory permissions

[Only researcher2 should be allowed to access the drafts directory and its contents. Current permissions from Step 3: drwx--x---user=rwx, group=--x, other=--- Problem: Group has execute permission. This allows group members to cd into the directory and access files if they know the names. The requirement says only researcher2 should have access. Linux command to modify:

A screenshot of a terminal window within a web browser. The browser's address bar shows a URL from qwiklabs.com. The terminal displays a series of commands and their outputs. First, 'chmod g-x drafts/' is executed. Then, 'ls -la' is run, showing a list of files and directories with their permissions, owner, group, size, and timestamps. The files listed are ., .., .project_x.txt, drafts, project_k.txt, project_m.txt, project_r.txt, and project_t.txt. The permissions for 'drafts' are drwxr-xr-x, and for the other files, they are -r--r-----. The terminal prompt is researcher2@929c8932blca:~/projects\$.

```
researcher2@929c8932blca:~/projects$ chmod g-x drafts/
researcher2@929c8932blca:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 17 12:46 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 17 13:24 ..
-r--r----- 1 researcher2 research_team 46 Jun 17 12:46 .project_x.txt
drwxr-xr-x 2 researcher2 research_team 4096 Jun 17 12:46 drafts
-rw-rw-r-- 1 researcher2 research_team 46 Jun 17 12:46 project_k.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jun 17 12:46 project_m.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jun 17 12:46 project_r.txt
-rw-rw-r-- 1 researcher2 research_team 46 Jun 17 12:46 project_t.txt
researcher2@929c8932blca:~/projects$
```

Removing execution from the group means only the owner researcher2 can list, create, or enter the drafts directory. "Other" already had no permissions. This ensures complete privacy for researcher2 drafts, aligning with the least-privilege principle for the research team.]

Summary

[I used ls -la to audit all files and directories, including the hidden .project_x.txt, and interpreted the 10-character permission strings to identify security violations. I then used chmod with symbolic notation to remove read access from "other" on project_k.txt, project_r.txt, and project_t.txt, remove write access from all users on the archived .project_x.txt while preserving group read, and remove group execute from the drafts/ directory. These changes ensure only authorized research team members can access project files and that researcher2 has exclusive access to private drafts, directly fulfilling the organization's security requirements.]